

GREEN UNIVERSITY OF BANGLADESH

Department of Artificial Intelligence and Data Science
B.Sc. in Artificial Intelligence and Data Science (ADS)

CareerBridge BD

A Job Portal for Fresh Graduates of Bangladesh
Software Engineering & Database System Project

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Table of Contents

Table of Contents	2
Abstract.....	4
1. Introduction	5
1.1 Background	5
1.2 Project Overview	5
1.3 Motivation	5
2. Problem Statement.....	6
3. Objectives	7
3.1 Primary Objectives	7
3.2 Secondary Objectives.....	7
4. Scope & Limitations	8
4.1 Scope.....	8
4.2 Limitations	8
5. Requirement Engineering	9
5.1 Functional Requirements.....	9
5.1.1 Graduate (Job Seeker)	9
5.1.2 Employer	9
5.1.3 Admin.....	9
5.2 Non-Functional Requirements.....	10
6. Use Case Diagram	11
6.1 Actors	11
6.2 Use Cases per Actor	11
6.3 Key Use Case: Apply for Job	11
7. System Architecture.....	12
7.1 Two-Stack Architecture Overview	12
7.2 Project Folder Structure	12
7.3 API Design (RESTful)	12
8. ER Diagram + Database Design.....	14
8.1 Entities & Relationships	14
8.2 Complete MySQL Schema.....	14
9. Wireframes / UI Design	15
9.1 Homepage.....	16
9.2 Graduate Dashboard.....	16
9.3 Job Listings Page	16
9.4 Employer Dashboard.....	16
10. Implementation & Tech Stack.....	17

11. GitHub Weekly Development Plan	18
12. Conclusion	19
13. References	20

Abstract

CareerBridge BD is a full-stack web-based job portal specifically designed to bridge the gap between fresh graduates of Bangladesh and employers who are actively looking for entry-level talent. In Bangladesh, thousands of students graduate each year from public and private universities, yet a significant portion faces extreme difficulty in finding relevant employment opportunities due to a lack of proper, localized platforms tailored to their needs.

This project proposes and implements a two-stack architecture web application React.js on the frontend and Node.js with Express on the backend supported by a MySQL relational database. The system allows fresh graduates to create profiles, upload CVs, browse job listings, and track application statuses. Employers can post jobs, manage applicants, and shortlist candidates. An admin panel governs platform integrity.

This documentation covers the complete initial phase of the project including requirement engineering, system architecture design, ER diagrams, use case and activity diagrams, and GitHub-based development plan.

1. Introduction

1.1 Background

Bangladesh produces over 300,000 university graduates annually. However, the job market remains highly competitive and poorly connected most job platforms are either generic, international, or too complex for fresh graduates. CareerBridge BD is built to directly address this pain point.

1.2 Project Overview

Project Name	CareerBridge BD — Job Portal for Fresh Graduates
Type	Two-Stack Full-Stack Web Application
Frontend	React.js
Backend	Node.js + Express.js
Database	MySQL
Target Users	Fresh Graduates, Employers, Platform Admin
Deployment	Localhost (Development) → GitHub Pages / Render (Optional)
Version Control	GitHub (Weekly Commits)

1.3 Motivation

When students graduate, many are unfamiliar with the job market, struggle to create professional CVs, and do not know how to approach employers. CareerBridge BD acts as a guided portal that helps fresh graduates present themselves professionally and connect with relevant entry-level opportunities in Bangladesh.

2. Problem Statement

The current job market ecosystem in Bangladesh suffers from several critical issues that specifically harm fresh graduates:

- Existing platforms (e.g., Bdjobs, LinkedIn) are general-purpose and do not cater specifically to fresh graduates with no prior work experience.
- Many job listings require 2–5 years of experience, making it nearly impossible for new graduates to apply.
- Fresh graduates lack professional CVs and portfolio guidance, reducing their visibility to recruiters.
- Employers have no efficient way to filter and find fresh, trainable talent from specific universities or fields.
- There is no centralized platform where fresh graduates can track their applications, get feedback, or schedule interviews.
- Geographic and informational barriers exist many opportunities remain limited to Dhaka and are not accessible to graduates from other districts.

CareerBridge BD directly solves these problems by providing a dedicated, structured, and accessible platform that gives fresh graduates a fair chance at the job market.

3. Objectives

3.1 Primary Objectives

1. Build a two-stack web application (React.js + Node.js) with MySQL database backend.
2. Design and implement a fully normalized relational database schema for the job portal.
3. Enable fresh graduates to register, build profiles, upload CVs, and apply for jobs.
4. Enable employers to post job listings, manage applications, and shortlist candidates.
5. Implement role-based authentication for three user types: Graduate, Employer, Admin.

3.2 Secondary Objectives

- Implement a job recommendation system based on graduate's skills and education.
 - Provide email/notification alerts for application status updates.
 - Build an admin dashboard to manage users, listings, and platform activity.
-

4. Scope & Limitations

4.1 Scope

The following modules are within the scope of this project:

Module	Description
User Management	Registration, Login, Profile Management for Graduates, Employers, Admin
Job Listings	Post, Edit, Delete, Search, Filter job advertisements
Application System	Apply for jobs, track status, employer review panel
CV Management	Upload CV (PDF), parse basic information into profile
Admin Panel	Manage users, moderate listings, view platform analytics
Notification System	Email alerts for application status changes

4.2 Limitations

- This project will not integrate real payment gateways (e.g., bKash, SSL Commerz) in the initial phase.
- No mobile application will be developed the platform is web-only.
- AI-powered resume scoring and video interviews are out of scope for this semester.
- Email notifications will be simulated using NodeMailer in development mode only.

5. Requirement Engineering

5.1 Functional Requirements

5.1.1 Graduate (Job Seeker)

Req. ID	Description
REQ-G01	Register with email, password, university, and graduation year
REQ-G02	Login and logout securely using JWT authentication
REQ-G03	Create and edit personal profile (skills, education, photo)
REQ-G04	Upload CV in PDF format
REQ-G05	Search and filter job listings by category, location, and type
REQ-G06	Apply for a job listing with one click
REQ-G07	Track all submitted applications and their statuses
REQ-G08	Receive email notification when application status changes

5.1.2 Employer

Req. ID	Description
REQ-E01	Register as a company with name, industry, location
REQ-E02	Login and manage company profile
REQ-E03	Post new job listings with title, description, requirements, deadline
REQ-E04	View all applicants for a specific job posting
REQ-E05	Shortlist, reject, or accept applicants
REQ-E06	Download applicant CV directly from the platform
REQ-E07	Edit or delete existing job postings

5.1.3 Admin

Req. ID	Description
REQ-A01	Login to admin dashboard
REQ-A02	View and manage all registered users (graduates and employers)
REQ-A03	Approve or reject employer registration requests

Req. ID	Description
REQ-A04	Remove inappropriate job listings
REQ-A05	View platform statistics (total users, active jobs, applications)

5.2 Non-Functional Requirements

NFR Type	Requirement
Performance	Page load time should be under 3 seconds on a standard connection
Security	Passwords must be hashed using bcrypt; JWT used for session management
Scalability	Database schema and API structure must support future feature additions
Usability	UI must be clean, responsive, and accessible on desktop browsers
Reliability	API endpoints must return proper HTTP status codes and error messages
Maintainability	Code must follow modular structure with clear folder conventions
Portability	Application must run on any system with Node.js and MySQL installed

6. Use Case Diagram

The use case diagram describes the interactions between the three primary actors Graduate, Employer, and Admin and the core system features of CareerBridge BD.

Note: Will be updated soon. The textual specification below fully describes the diagram structure.

6.1 Actors

- Graduate (Job Seeker) — Primary user who searches and applies for jobs
- Employer (Company) — Posts jobs and manages applications
- Admin — Manages platform integrity and user verification
- System — Handles automated background actions (email, auto-expire)

6.2 Use Cases per Actor

Actor	Use Cases
Graduate	Register/Login, Build Profile, Upload CV, Search Jobs, Apply for Job, Track Applications, Receive Notifications
Employer	Register/Login, Post Job, Manage Listings, View Applicants, Shortlist/Reject Candidates, Download CV
Admin	Login to Dashboard, Manage Users, Approve Employers, Remove Listings, View Analytics
System (<<system>>)	Send Email on Status Change, Auto-expire Job Listings, Generate Application ID

6.3 Key Use Case: Apply for Job

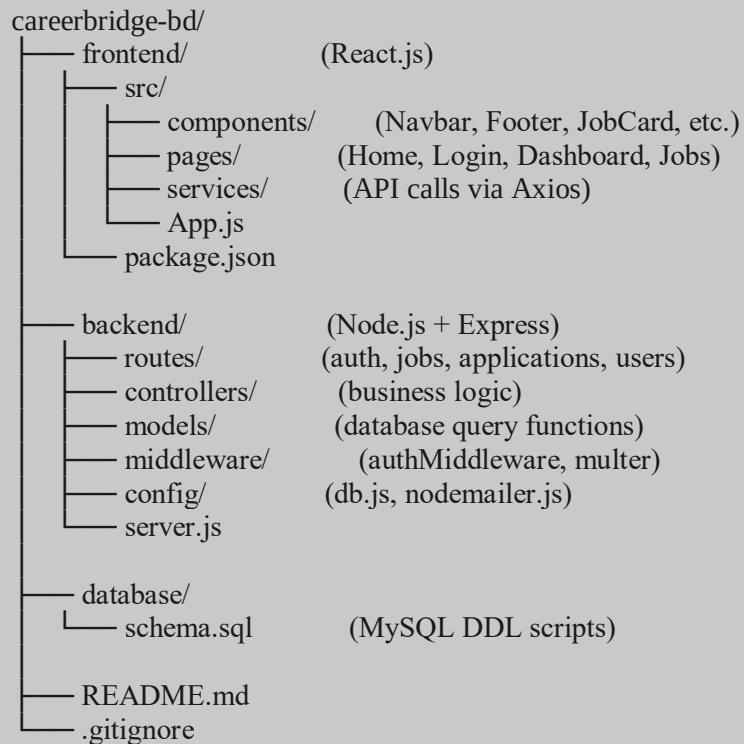
Field	Details
Use Case Name	Apply for Job
Actor	Graduate
Precondition	Graduate must be logged in and have an uploaded CV
Main Flow	1. Graduate searches for a job 2. Clicks 'Apply' 3. System checks profile completeness 4. Application stored in database 5. Employer notified
Alternate Flow	If CV is missing, system prompts graduate to upload CV first
Postcondition	Application appears in graduate's tracking list with status 'Pending'

7. System Architecture

7.1 Two-Stack Architecture Overview

Layer	Technology & Responsibility
Presentation Layer	React.js - Handles UI, routing, state management (React Router, Axios, Tailwind CSS)
API Layer	Node.js + Express.js - RESTful API endpoints, middleware, authentication
Data Layer	MySQL - Relational database for persistent data storage
Authentication	JWT (JSON Web Tokens) + bcrypt password hashing
File Storage	Multer middleware for CV uploads (stored in /uploads folder)
Email Service	NodeMailer (SMTP) for notification emails

7.2 Project Folder Structure



7.3 API Design (RESTful)

Endpoint	Description
POST /api/auth/register	Register new user (graduate or employer)
POST /api/auth/login	Login and receive JWT token
GET /api/jobs	Get all active job listings (paginated)
POST /api/jobs	Employer posts new job listing
GET /api/jobs/:id	Get single job details
POST /api/applications	Graduate applies for a job
GET /api/applications/mine	Graduate views own applications
PATCH /api/applications/:id	Employer updates application status
GET /api/admin/users	Admin views all users
PATCH /api/admin/employers/:id/verify	Admin approves/rejects an employer.

8. ER Diagram + Database Design

Note: Will be updated soon

8.1 Entities & Relationships

Table	Purpose
users	Stores all users (graduates, employers, admin) with role field
graduate_profiles	Extended profile for graduates — skills, education, CV path
employer_profiles	Company details for employer accounts
job_listings	All job posts created by employers
applications	Junction table — connects graduates to job listings
notifications	Stores notification messages for users

8.2 Complete MySQL Schema

```
-- TABLE: users
CREATE TABLE users (
  id INT PRIMARY KEY AUTO_INCREMENT,
  name VARCHAR(100) NOT NULL,
  email VARCHAR(150) UNIQUE NOT NULL,
  password VARCHAR(255) NOT NULL,
  role ENUM('graduate', 'employer', 'admin') NOT NULL,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- TABLE: graduate_profiles
CREATE TABLE graduate_profiles (
  id INT PRIMARY KEY AUTO_INCREMENT,
  user_id INT UNIQUE NOT NULL,
  university VARCHAR(150),
  graduation_year YEAR,
  skills TEXT,
  bio TEXT,
  cv_path VARCHAR(255),
  FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE CASCADE
);

-- TABLE: employer_profiles
CREATE TABLE employer_profiles (
  id INT PRIMARY KEY AUTO_INCREMENT,
  user_id INT UNIQUE NOT NULL,
  company_name VARCHAR(150) NOT NULL,
```

```

industry VARCHAR(100),
location VARCHAR(100),
website VARCHAR(200),
is_verified BOOLEAN DEFAULT FALSE,
FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE CASCADE
);

-- TABLE: job_listings
CREATE TABLE job_listings (
  id INT PRIMARY KEY AUTO_INCREMENT,
  employer_id INT NOT NULL,
  title VARCHAR(200) NOT NULL,
  description TEXT,
  requirements TEXT,
  location VARCHAR(100),
  job_type ENUM('full-time', 'part-time', 'internship', 'remote'),
  salary_range VARCHAR(100),
  deadline DATE,
  status ENUM('active', 'closed') DEFAULT 'active',
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  FOREIGN KEY (employer_id) REFERENCES users(id) ON DELETE CASCADE
);

-- TABLE: applications
CREATE TABLE applications (
  id INT PRIMARY KEY AUTO_INCREMENT,
  graduate_id INT NOT NULL,
  job_id INT NOT NULL,
  status ENUM('pending', 'shortlisted', 'rejected', 'hired') DEFAULT 'pending',
  applied_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  UNIQUE KEY unique_application (graduate_id, job_id),
  FOREIGN KEY (graduate_id) REFERENCES users(id) ON DELETE CASCADE,
  FOREIGN KEY (job_id) REFERENCES job_listings(id) ON DELETE CASCADE
);

-- TABLE: notifications
CREATE TABLE notifications (
  id INT PRIMARY KEY AUTO_INCREMENT,
  user_id INT NOT NULL,
  message TEXT,
  is_read BOOLEAN DEFAULT FALSE,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE CASCADE
);

```

9. Wireframes / UI Design

Note: Will be updates soon. The structural layout for each page is described below.

9.1 Homepage

Section	Content
Header	Logo (CareerBridge BD), Navigation (Home, Jobs, Login, Register)
Hero Section	Headline text, Search bar (Job title / Location), 'Find Jobs' CTA button
Stats Bar	Total Jobs: XXX Total Companies: XX Fresh Graduates Hired: XXX
Featured Jobs	Grid of 6 recent job cards with title, company, location, type badge
Footer	Links, contact info, social media icons

9.2 Graduate Dashboard

Section	Content
Sidebar	Profile Photo, Name, CV Upload button, Navigation links
Main Area	Profile Completeness Bar, Recent Applications Table, Recommended Jobs
Application Table	Job Title Company Applied Date Status (color-coded badge)

9.3 Job Listings Page

Section	Content
Left Sidebar	Filter panel: Job Type, Location, Industry, Salary Range
Main Grid	Job cards — Title, Company Logo, Location, Type, Posted Date, 'Apply' button
Pagination	Page 1, 2, 3... controls at the bottom

9.4 Employer Dashboard

Section	Content
Top Cards	Total Active Jobs Total Applications Received Shortlisted Candidates
My Listings	Table of posted jobs with Edit / View Applicants / Close buttons
Recent Applicants	Latest 5 applications across all jobs with quick action buttons

10. Implementation & Tech Stack

Technology	Tool	Reason
Frontend Framework	React.js 18	Component-based, fast re-rendering, wide community support
Routing	React Router v6	Client-side navigation without page reloads
HTTP Client	Axios	Promise-based API calls from React to Node backend
Styling	Tailwind CSS	Utility-first CSS framework for rapid UI development
Backend Framework	Node.js + Express.js	Lightweight, async-friendly RESTful API server
Authentication	JWT + bcrypt	Stateless auth tokens + secure password hashing
Database	MySQL 8.0	Reliable relational DB with ACID compliance
ORM / Query	mysql2 package	Direct SQL queries with connection pooling
File Upload	Multer	Middleware for handling CV (PDF) file uploads
Email	NodeMailer	SMTP-based email notifications
Version Control	Git + GitHub	Weekly commits, branch strategy, README documentation
IDE	VS Code	Code editing with ESLint, Prettier extensions

11. GitHub Weekly Development Plan

The following 9-week plan outlines week-by-week development milestones. Each week ends with a GitHub commit.

Week	Tasks	Deliverables	GitHub Commit
Week 1	Project setup, GitHub repo init, folder structure, README, documentation draft	Initial Documentation	Initial commit: project setup + docs
Week 2	MySQL schema creation, ER Diagram finalization, seed data scripts	database/schema.sql, ER Diagram image	feat: database schema + ER diagram
Week 3	Backend: Express server setup, auth routes (register/login), JWT middleware	Auth API endpoints working	feat: backend auth system
Week 4	Backend: Job listing routes (CRUD), application routes, middleware	Jobs & Application API complete	feat: jobs and applications API
Week 5	Frontend: React app setup, routing, Login & Register pages, Axios integration	Auth UI working end-to-end	feat: frontend auth pages
Week 6	Frontend: Homepage, Job Listings page, Job Detail page, Apply button	Core job browsing UI complete	feat: job listing frontend
Week 7	Frontend: Graduate Dashboard, Application Tracker, Employer Dashboard, Admin Panel	All dashboards complete	feat: dashboards and admin panel
Week 8	CV upload (Multer), Email notifications (NodeMailer), testing & bug fixing	File upload + email alerts working	feat: CV upload and notifications
Week 9	Final testing, UI polish, README update, deployment (optional), demo video	Complete project + final documentation	chore: final polish and submission

12. Conclusion

CareerBridge BD represents a meaningful and technically rigorous response to a real problem faced by thousands of fresh graduates in Bangladesh. By building a two-stack web application with React.js on the frontend, Node.js on the backend, and MySQL as the relational database, this project covers all aspects of full-stack software engineering while adhering to professional practices.

The initial documentation presented here demonstrates a thorough understanding of the project requirements, system design, database architecture, and development planning. The 9-week GitHub-based development plan ensures the project progresses systematically with visible, reviewable deliverables each week.

Upon completion, CareerBridge BD will stand as a complete, deployable, professional-grade web application that demonstrates strong competencies in Software Engineering, Database Design, React development, and Node.js backend architecture.

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